The induced transformation on a symmetric subinterval satisfies the same conjugacy equation

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Editorial note. The counted statement is the author's labelled lemma environment at source0.tex lines 626-628; its proof environment opens at line 630, the line after the blank separator, so the pairing is by adjacency and the retained proof range 630-646 is the whole environment.

Editorial note. Two statements are retained as context because the counted statement and proof refer to them: the conjugacy equation eq: standardconjugacy (source lines 463-465), which the statement itself displays, and the local reflection lemma (source lines 596-602), which the proof cites at source line 639.

Editorial note. This source is new to the material-authoring work and its preamble is built from the author's own macro and theorem block (source lines 8-45) rather than from a prepared adapter.

Editorial note. This source has no label-number dump and the retained counter values are left empty, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label.

Editorial note. r002 note: r001 failed with an undefined reference to eq:generalized_conjugacy, which the counted proof applies at source lines 631 and 641; r002 adds exactly that numbered identity as a further context and changes nothing else.

Editorial scope. One original result of the pinned publisher article, counted once: the induced transformation on a symmetric subinterval satisfies the conjugacy equation (author label lem:symmetric_condition).

Allowed standard prerequisites (editorial).

- Induced transformation. The induced transformation on a subinterval is defined by the first return time of its points to the subinterval.. It is the transformation about which the statement is made.
- Conjugacy equation. For a symmetric transformation the reflection of the whole interval conjugates it to its inverse off the discontinuities.. It is the equation the induced transformation is asserted to satisfy.
- Symmetry of subintervals. A subinterval is either alpha-symmetric for a continuity interval or symmetric about the middle of the whole interval.. It is the case distinction along which the proof runs.

$$\mathcal{I}_I \circ T^n(x) = T^{-n} \circ \mathcal{I}_I(x) \quad \text{if } x \neq T^{-i}(\partial I_\alpha) \text{ for any } \alpha \in \mathcal{A} \text{ and any } i \text{ such that } 0 \leq i < n. \quad (1)$$

$$\mathcal{I}_I \circ T(x) = T^{-1} \circ \mathcal{I}_I(x) \quad \text{if } x \neq \partial I_\alpha \text{ for any } \alpha \in \mathcal{A}. \quad (2)$$

Lemma 0.1. *Let $T : I \to I$ be a symmetric IET and $J \subseteq I$ be an α-symmetric interval, for some $\alpha \in \mathcal{A}$. Then $\mathcal{I}_I \circ T(\mathring{J}) = \mathring{J}$. Moreover,*

$$\mathcal{I}_J = \mathcal{I}_I \circ T|_{\mathring{J}},$$

that is, $\mathcal{I}_I \circ T|_{\mathring{J}}$ is the symmetric reflection on J.

Lemma 0.2. *Let $T : I \to I$ be a symmetric IET and $J \subseteq I$ be a symmetric interval. Then T_J satisfies* (2)*.*

Proof. Assume first that J is α-symmetric for some $\alpha \in \mathcal{A}$. Let $x \in J$ be not an endpoint of an interval exchanged by T_J and let $h \in \mathbb{N}$ be such that $T_J(x) = T^h(x)$. Notice that since x is not an endpoint of the exchanged intervals, by (1) applied to T,

$$(\mathcal{I}_I \circ T) \circ T_J(x) = (\mathcal{I}_I \circ T) \circ T^h(x) = T^{-h} \circ (\mathcal{I}_I \circ T)(x).$$

Thus, to prove (2), it suffices to notice that

$$T_J^{-1} \circ (\mathcal{I}_I \circ T)(x) = T^{-h} \circ (\mathcal{I}_I \circ T)(x).$$

Indeed, by Lemma 0.1, for any $n \geq 1$, $T^n(x) \in J$ if and only if $(\mathcal{I}_I \circ T) \circ T^n(x) \in J$. Thus the equation above follows since $T_J(x) = T^h(x)$ and $T^{-h} \circ (\mathcal{I}_I \circ T)(x) = (\mathcal{I}_I \circ T) \circ T^h(x)$.

Assume now that J is $1/2$-symmetric. Let $x \in J$ be not an endpoint of an interval exchanged by T_J and let $h \in \mathbb{N}$ be such that $T_J(x) = T^h(x)$. Again by (1) we get

$$\mathcal{I}_I \circ T_J(x) = \mathcal{I}_I \circ T^h(x) = T^{-h} \circ \mathcal{I}_I(x)$$

and, similarly to the previous case, to show (2) it is sufficient to notice that $T_J^{-1} \circ \mathcal{I}_I(x) = T^{-h} \circ \mathcal{I}_I(x)$. □